

CAMBRIDGE INTERNATIONAL AS & A LEVEL CHEMISTRY (9701)

Topic 1 — Atomic Structure (Paper 1 Multiple Choice)

Paper 1 (Multiple Choice) Topical Past Paper Pack · 110 Questions & Detailed Solutions

CANDIDATE	Urwah
SYLLABUS	Cambridge International AS Level Chemistry (9701)
COMPONENT	Paper 1: Multiple Choice (1 Hour 15 Minutes Exam Standard)
TOPIC COVERAGE	1.1 Particles & Radii · 1.2 Isotopes · 1.3 Orbitals & Configurations · 1.4 Ionisation Energies · High-Frequency Repeats
TOTAL VOLUME	110 Authentic Multiple Choice Questions (110 Marks · 1 Mark Each)

HOW TO USE THIS PAPER 1 MCQ PACK

1. Timing & Strategy: Spend approximately 1 minute to 1 minute 15 seconds per question under timed test conditions.
2. Answering: For each question, select the one correct option from A, B, C, or D. Apply the elimination method to rule out obvious distractors.
3. Authentic Exam Codes: Every question features authentic Cambridge reference codes (e.g. 9701/12/M/J/23) for cross-study verification.
4. Quick-Check Answer Grid: Use the 110-question matrix immediately following the questions for rapid scoring in under 60 seconds.
5. Comprehensive Explanations: Review the detailed examiner rationale and distractor analysis at the back to eliminate conceptual errors.

SYLLABUS SUBTOPICS COVERED IN THIS PACK

1.1 Particles & Atomic Radii

— Fundamental particles, charges, masses, deflection in electric fields, nuclear charge, shielding, and periodic trends

1.2 Isotopes & Relative Masses

— Definition of isotopes, mass spectrometry principles, calculating relative atomic mass A_r , and polyatomic isotopes

1.3 Electrons & Orbitals

— Principal shells (n), subshells (s, p, d), orbital shapes, Aufbau principle, Hund's rule, Pauli exclusion, transition metals

1.4 Ionisation Energies

— First and successive ionisation energies, periodic trends across Period 3, Group 13/16 anomalies (Al vs Mg, S vs P)

High-Frequency Core Repeats

— The 10 most frequently examined Cambridge Paper 1 questions on Atomic Structure (Q101 – Q110).

SUBTOPIC 1.1 — PARTICLES IN THE ATOM & ATOMIC RADIUS (Q1 – Q25)

1 [EASY] Fundamental Particles and Mass — 9701/12/M/J/23/Q1

[1]

Which statement regarding the fundamental subatomic particles of an atom is correct?

- [A] Protons and electrons have equal masses but opposite electrical charges.
- [B] The mass of an electron is approximately $1/1836$ of the mass of a proton.
- [C] Neutrons have a relative charge of -1 and reside outside the nucleus.
- [D] The nucleus contains protons and electrons held by electrostatic attraction.

2 [HARD] Deflection of Subatomic Particles in an Electric Field — 9701/11/O/N/23/Q1

[1]

Beams of protons, neutrons, and electrons travel at equal velocities into an electric field between two oppositely charged plates. Fig. 2.1 illustrates the path taken by the particles.

Which statement correctly describes the deflection of these particles?

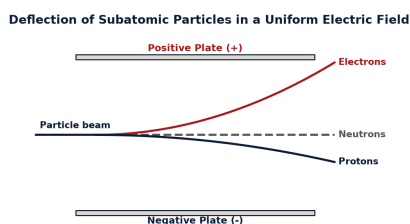


Fig. 2.1: Paths of subatomic particle beams passing through an electric field.

- [A] Protons and electrons are deflected by equal angles towards opposite plates because they carry equal magnitudes of charge.
- [B] Neutrons are deflected towards the negative plate because they are massive particles.
- [C] Protons are deflected towards the positive plate because they carry a positive charge.
- [D] Electrons are deflected towards the positive plate with a much greater angle of deflection than protons towards the negative plate.

3 [EASY] Deducing Subatomic Particle Numbers in Ions — 9701/13/M/J/22/Q1

[1]

An ion of an element is represented as ${}^{59}_{28}\text{Ni}^{2+}$. How many protons, neutrons, and electrons are present in this ion?

- [A] 28 protons, 31 neutrons, 26 electrons
- [B] 28 protons, 31 neutrons, 28 electrons
- [C] 28 protons, 59 neutrons, 26 electrons
- [D] 31 protons, 28 neutrons, 26 electrons

4 [HARD] Comparison of Deflection Angles of Gaseous Ions — 9701/12/F/M/24/Q1

[1]

Beams of the following gaseous ions are accelerated to the same velocity and passed through identical electric fields:

- 1: H^+
- 2: 2H^+
- 3: 4He^{2+}
- 4: 12C^{2+}

Which two ions experience exactly the same angle of deflection?

- [A] H^+ and 2H^+
- [B] 4He^{2+} and 12C^{2+}
- [C] 2H^+ and 4He^{2+}
- [D] H^+ and 4He^{2+}

5 [EASY] Atomic Radius Trend Across Period 3 — 9701/11/M/J/23/Q2

[1]

Why does atomic radius decrease across Period 3 from sodium to chlorine?

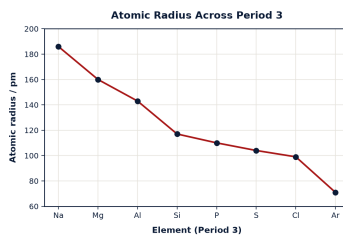


Fig. 5.1: Variation of atomic radius across Period 3 elements.

- [A] Nuclear charge increases while shielding remains approximately constant, pulling electrons closer.
- [B] The number of occupied principal quantum shells increases across the period.
- [C] Electron-electron repulsions in the outer shell significantly increase the screening effect.
- [D] The electronegativity of the elements decreases from left to right.

6 [HARD] Comparing Radii of Isoelectronic Species — 9701/12/O/N/23/Q2

[1]

Consider the following four isoelectronic species:

N^{3-} , O^{2-} , F^- , Na^+

Which list places these species in order of increasing ionic radius (smallest to largest)?

- [A] $\text{N}^{3-} < \text{O}^{2-} < \text{F}^- < \text{Na}^+$
- [B] $\text{F}^- < \text{O}^{2-} < \text{N}^{3-} < \text{Na}^+$
- [C] $\text{Na}^+ < \text{N}^{3-} < \text{O}^{2-} < \text{F}^-$
- [D] $\text{Na}^+ < \text{F}^- < \text{O}^{2-} < \text{N}^{3-}$

7 [EASY] Comparison of Atomic vs Ionic Radius — 9701/13/O/N/22/Q2

[1]

Which statement correctly compares an atom with its corresponding ion?

- [A] A sodium atom has a smaller radius than a sodium ion, Na^+ .
- [B] A chlorine atom has a smaller radius than a chloride ion, Cl^- .
- [C] A magnesium ion, Mg^{2+} , has a larger radius than a magnesium atom.
- [D] An oxide ion, O^{2-} , has a smaller radius than an oxygen atom.

8 [HARD] Effective Nuclear Charge Calculation — 9701/12/M/J/22/Q2

[1]

Which element has valence electrons that experience the greatest effective nuclear charge (Z_{eff})?

- [A] Silicon ($Z = 14$)
- [B] Magnesium ($Z = 12$)
- [C] Chlorine ($Z = 17$)
- [D] Sodium ($Z = 11$)

9 [EASY] Subatomic Particles in Tripositive Cation — 9701/11/F/M/23/Q1

[1]

An ion X^{3+} contains 23 electrons and 30 neutrons. What is the nucleon (mass) number of element X?

- [A] 53
- [B] 56
- [C] 50
- [D] 26

10 [HARD] Deflection of Doubly Charged Oxygen Anion — 9701/12/O/N/21/Q1**[1]**

A beam containing 16O^{2-} ions enters an electric field where a beam of 1H^{+} ions is deflected by an angle of $+16.0^\circ$ towards the negative plate.

What is the angle and direction of deflection for the 16O^{2-} beam?

- [A] -2.0° (towards the positive plate)
- [B] $+2.0^\circ$ (towards the negative plate)
- [C] -4.0° (towards the positive plate)
- [D] -8.0° (towards the positive plate)

11 [EASY] Identifying Elements with Same Number of Neutrons — 9701/13/M/J/24/Q1**[1]**

Which pair of atomic species contains exactly the same number of neutrons?

- [A] $^{14}_6\text{C}$ and $^{14}_7\text{N}$
- [B] $^{40}_{18}\text{Ar}$ and $^{40}_{20}\text{Ca}$
- [C] $^{23}_{11}\text{Na}$ and $^{24}_{12}\text{Mg}$
- [D] $^{31}_{15}\text{P}$ and $^{32}_{16}\text{S}$

12 [EASY] Comparison of Group 2 Atomic Radii — 9701/11/M/J/22/Q2**[1]**

Why does the atomic radius increase down Group 2 from beryllium to barium?

- [A] The nuclear charge decreases down the group.
- [B] The outer s-electrons experience less total electrostatic pull because mass increases.
- [C] Each successive element has an additional principal quantum shell of electrons.
- [D] The number of neutrons in the nucleus repels the outer electron cloud.

13 [EASY] Identifying Isoelectronic Ions — 9701/12/M/J/21/Q2**[1]**

Which set of ions all have the same electronic configuration as argon (Ar)?

- [A] K^{+} , Ca^{2+} , Cl^{-} , S^{2-}
- [B] Na^{+} , Mg^{2+} , F^{-} , O^{2-}
- [C] K^{+} , Ca^{2+} , Br^{-} , Se^{2-}
- [D] Sc^{3+} , Ti^{4+} , P^{3-} , O^{2-}

14 [HARD] Calculating Relative Particle Charge in Deuterium Ion — 9701/13/O/N/23/Q1**[1]**

A deuterium hydride ion, $[\text{2H}]^{-}$, is formed by adding an electron to a neutral deuterium atom.

What is the total number of subatomic particles (protons + neutrons + electrons) in one $[\text{2H}]^{-}$ ion?

- [A] 3
- [B] 5
- [C] 4
- [D] 2

15 [HARD] Trend in Ionic Radius Across Period 3 Cations — 9701/11/O/N/22/Q2**[1]**

Consider the cations Na^{+} , Mg^{2+} , and Al^{3+} . Which statement correctly explains why Al^{3+} has the smallest ionic radius?

- [A] Al^{3+} has three valence shells whereas Na^{+} has only two.
- [B] Al^{3+} has the greatest nuclear charge (+13) attracting the same number of electrons (10).
- [C] The shielding effect in Al^{3+} is significantly greater than in Na^{+} .
- [D] Al^{3+} has fewer protons than neutrons in its nucleus.

16 [EASY] Particles in the Nucleus of an Isotope — 9701/12/F/M/22/Q1

[1]

An atom of element Y has 35 protons, 45 neutrons, and 35 electrons. Which symbol correctly represents this atom?

- [A] $^{45}_{35}\text{Br}$
- [B] $^{80}_{45}\text{Rh}$
- [C] $^{70}_{35}\text{Br}$
- [D] $^{80}_{35}\text{Br}$

17 [HARD] Deflection of Isotopes in an Electric Field — 9701/13/M/J/23/Q2

[1]

Beams of singly charged ions of carbon isotopes, $^{12}\text{C}^+$ and $^{14}\text{C}^+$, pass through the same electric field with the same velocity. Which statement is correct?

- [A] $^{14}\text{C}^+$ is deflected more than $^{12}\text{C}^+$ because it has more neutrons.
- [B] Both ions are deflected by the same angle because both have a +1 charge.
- [C] $^{12}\text{C}^+$ is deflected more than $^{14}\text{C}^+$ because it has a smaller mass.
- [D] $^{12}\text{C}^+$ is deflected towards the positive plate and $^{14}\text{C}^+$ towards the negative plate.

18 [HARD] Number of Unpaired Electrons in Transition Metal Cation — 9701/11/M/J/24/Q2

[1]

How many unpaired electrons are present in a gaseous Fe^{3+} ion in its ground state?

- [A] 5
- [B] 4
- [C] 3
- [D] 1

19 [HARD] Comparison of Anion Radii Across Period 3 — 9701/12/M/J/23/Q2

[1]

Why is the ionic radius of P^{3-} (phosphide) larger than that of Cl^- (chloride)?

- [A] P^{3-} has electrons in the fourth principal quantum shell.
- [B] Cl^- has more shielding than P^{3-} .
- [C] The phosphorus nucleus contains fewer neutrons than the chlorine nucleus.
- [D] P^{3-} has a lower nuclear charge (+15) attracting the same number of electrons (18) as Cl^- (+17).

20 [EASY] Ratio of Electron to Proton Mass — 9701/11/O/N/21/Q2

[1]

An atom contains 6 protons, 6 neutrons, and 6 electrons. What percentage of the total mass of the atom is contributed by the electrons?

- [A] Approximately 1.0%
- [B] Less than 0.05%
- [C] Approximately 5.0%
- [D] Approximately 33.3%

21 [EASY] Nuclear Composition of Uranium-235 and 238 — 9701/13/O/N/21/Q1

[1]

Which statement correctly compares an atom of $^{235}_{92}\text{U}$ with an atom of $^{238}_{92}\text{U}$?

- [A] $^{238}_{92}\text{U}$ contains 3 more protons than $^{235}_{92}\text{U}$.
- [B] $^{238}_{92}\text{U}$ contains 3 more neutrons than $^{235}_{92}\text{U}$.
- [C] $^{238}_{92}\text{U}$ contains 3 more electrons than $^{235}_{92}\text{U}$.
- [D] $^{238}_{92}\text{U}$ has a different chemical reactivity than $^{235}_{92}\text{U}$.

22 [HARD] Atomic Radius Discontinuity from Group 2 to Group 13 — 9701/12/M/J/24/Q1**[1]**

Across Period 3, atomic radius decreases from Mg to Al, but the decrease is smaller than from Na to Mg. What explains this effect?

- [A] Magnesium has a higher electronegativity than aluminum.
- [B] Aluminum has a smaller nuclear charge than magnesium.
- [C] The 3p electron in aluminum causes pairing repulsion with the 3s electrons.
- [D] In aluminum, the incoming electron enters the 3p subshell, which is further from the nucleus and partially shielded by the 3s² electrons.

23 [EASY] Identifying Isoelectronic Noble Gas — 9701/11/F/M/24/Q1**[1]**

Which ion has the same electron configuration as neon?

- [A] Mg²⁺
- [B] Ca²⁺
- [C] Cl⁻
- [D] K⁺

24 [HARD] Deflection of Alpha and Beta Particles — 9701/12/O/N/23/Q1**[1]**

In an electric field, alpha particles (4He²⁺) and beta particles (electrons, e⁻) are deflected in opposite directions. Why is the angle of deflection of a beta particle much greater than that of an alpha particle?

- [A] Beta particles carry a charge twice as large as alpha particles.
- [B] Alpha particles travel at the speed of light.
- [C] The charge-to-mass ratio of a beta particle is approximately 3670 times greater than that of an alpha particle.
- [D] Beta particles are neutral and unaffected by gravity.

25 [EASY] Calculating Relative Atomic Mass from Subatomic Numbers — 9701/13/M/J/22/Q2**[1]**

An imaginary atom of element M has 5 protons, 6 neutrons, and 5 electrons. What is its approximate mass relative to the unified atomic mass unit (u)?

- [A] 11
- [B] 5
- [C] 6
- [D] 16

SUBTOPIC 1.2 — ISOTOPES & RELATIVE ATOMIC MASS (Q26 – Q50)**26 [EASY]** Definition of Isotopes — 9701/12/M/J/23/Q3**[1]**

Which statement correctly defines isotopes of an element?

- [A] Atoms of different elements having the same nucleon number.
- [B] Molecules of the same formula but different structural arrangements of atoms.
- [C] Ions of the same element with different numbers of electrons.
- [D] Atoms of the same element having the same number of protons but different numbers of neutrons.

27 [EASY] Chemical Properties of Isotopes — 9701/11/O/N/23/Q3**[1]**

Why do isotopes of the same element exhibit identical chemical reactions?

- [A] They have identical numbers of neutrons in their atomic nuclei.
- [B] They have identical electronic configurations and the same number of valence electrons.
- [C] They have identical relative atomic masses.
- [D] Their nuclear binding energies are identical.

28 [HARD] Physical Properties of Isotopes — 9701/13/F/M/23/Q2

[1]

Which physical property differs between H_2O (ordinary water) and D_2O (heavy water, D_2O)?

- [A] The oxidation state of oxygen
- [B] The pH of pure neutral liquid
- [C] Density and boiling point
- [D] The formula of the salt formed with sodium hydroxide

29 [EASY] Calculating Relative Atomic Mass of Boron — 9701/12/M/J/22/Q3

[1]

Naturally occurring boron consists of 19.9% ^{10}B and 80.1% ^{11}B .

What is the relative atomic mass, A_r , of boron to two decimal places?

- [A] 10.50
- [B] 10.80
- [C] 10.20
- [D] 10.95

30 [HARD] Deducing Isotopic Abundances from A_r — 9701/11/M/J/23/Q3

[1]

Copper has a relative atomic mass of 63.55 and consists of only two isotopes, ^{63}Cu and ^{65}Cu .

What is the percentage abundance of ^{63}Cu ?

- [A] 72.5%
- [B] 27.5%
- [C] 63.5%
- [D] 55.0%

31 [HARD] Mass Spectrometry of Zirconium — 9701/12/O/N/22/Q3

[1]

The mass spectrum of zirconium consists of five isotopic peaks as shown in Fig. 31.1:

^{90}Zr (51.5%), ^{91}Zr (11.2%), ^{92}Zr (17.1%), ^{94}Zr (17.4%), ^{96}Zr (2.8%).

What is the relative atomic mass of this sample of zirconium?

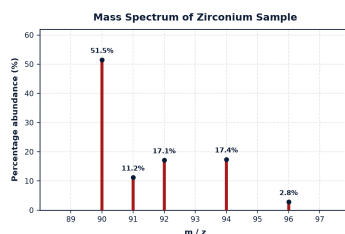


Fig. 31.1: Mass spectrum of zirconium showing five isotopic peaks.

- [A] 92.50
- [B] 90.00
- [C] 94.20
- [D] 91.32

32 [HARD] Diatomic Chlorine Mass Spectrum Peaks — 9701/13/O/N/23/Q2

[1]

Naturally occurring chlorine consists of ^{35}Cl and ^{37}Cl in a 3:1 ratio.

When diatomic chlorine, Cl_2 , is analyzed by mass spectrometry, how many peaks are observed in the molecular ion region, and what is their relative intensity ratio?

- [A] 2 peaks with ratio 3 : 1 (at m/z 70, 72)
- [B] 3 peaks with ratio 3 : 2 : 1 (at m/z 70, 72, 74)
- [C] 3 peaks with ratio 9 : 6 : 1 (at m/z 70, 72, 74)
- [D] 4 peaks with ratio 1 : 2 : 2 : 1 (at m/z 70, 71, 72, 73)

33 [HARD] Diatomic Bromine Mass Spectrum Peaks — 9701/11/M/J/22/Q3**[1]**

Bromine has two isotopes, ^{79}Br and ^{81}Br , with approximately equal abundances (50% each). What is the expected ratio of peak heights in the molecular ion region of Br_2^+ ?

- [A] 1 : 2 : 1 at m/z 158, 160, 162
- [B] 1 : 1 : 1 at m/z 158, 160, 162
- [C] 1 : 1 at m/z 158, 162
- [D] 3 : 1 at m/z 158, 160

34 [EASY] Definition of Relative Atomic Mass — 9701/12/F/M/23/Q1**[1]**

Which statement provides the precise definition of relative atomic mass, A_r ?

- [A] The mass of an atom of an element compared to the mass of a hydrogen atom.
- [B] The sum of the number of protons and neutrons in the nucleus of the most abundant isotope.
- [C] The weighted average mass of an atom of an element compared to $1/12$ the mass of an atom of carbon-12.
- [D] The mass of one mole of atoms of the element measured in grams.

35 [HARD] Relative Isotopic Mass vs Relative Atomic Mass — 9701/13/M/J/21/Q2**[1]**

Why is the relative isotopic mass of ^{35}Cl exactly 34.97, whereas the relative atomic mass of chlorine is 35.45?

- [A] Relative isotopic mass includes the mass of electrons while relative atomic mass does not.
- [B] Relative atomic mass is a weighted average of all naturally occurring isotopes, whereas relative isotopic mass refers to a single specific isotope.
- [C] The mass of chlorine changes when it forms chemical compounds.
- [D] Chlorine undergoes radioactive decay in the mass spectrometer.

36 [EASY] Calculating Relative Atomic Mass of Magnesium — 9701/11/O/N/21/Q3**[1]**

A sample of magnesium contains three isotopes:

^{24}Mg (79.0%), ^{25}Mg (10.0%), ^{26}Mg (11.0%)

What is the relative atomic mass of magnesium in this sample?

- [A] 24.00
- [B] 25.00
- [C] 24.50
- [D] 24.32

37 [HARD] Deducing Abundance of a Third Isotope — 9701/12/M/J/24/Q2**[1]**

Element E has three isotopes: ^{28}E , ^{29}E , and ^{30}E . The abundance of ^{28}E is 92.2% and the relative atomic mass of E is 28.09.

What is the percentage abundance of ^{30}E ?

- [A] 4.7%
- [B] 7.8%
- [C] 3.1%
- [D] 1.5%

38 [EASY] Isotopic Composition of Neon in Mass Spectrum — 9701/13/O/N/22/Q1**[1]**

Neon has two major isotopes, ^{20}Ne and ^{22}Ne . In a mass spectrometer, the peak at $m/z = 20$ has a height of 90 mm and the peak at $m/z = 22$ has a height of 10 mm.

What is the relative atomic mass of neon from these data?

- [A] 20.2
- [B] 20.0
- [C] 21.0
- [D] 21.8

39 [EASY] Identifying Elements with Same Number of Nucleons — 9701/11/F/M/22/Q1

[1]

Which two species have the same nucleon number (isobars)?

- [A] $^{12}_{6}\text{C}$ and $^{14}_{6}\text{C}$
- [B] $^{16}_{8}\text{O}$ and $^{18}_{8}\text{O}$
- [C] $^{35}_{17}\text{Cl}$ and $^{37}_{17}\text{Cl}$
- [D] $^{14}_{6}\text{C}$ and $^{14}_{7}\text{N}$

40 [HARD] Carbon-14 Radioactive Decay Product — 9701/12/M/J/21/Q1

[1]

Carbon-14 ($^{14}_{6}\text{C}$) undergoes beta decay to form an atom of nitrogen.
Which nuclear equation correctly describes this process?

- [A] $^{14}_{6}\text{C} \rightarrow ^{13}_{7}\text{N} + ^1_0\text{n}$
- [B] $^{14}_{6}\text{C} \rightarrow ^{14}_{7}\text{N} + ^0_{-1}\text{e}$
- [C] $^{14}_{6}\text{C} \rightarrow ^{10}_{4}\text{Be} + ^4_2\text{He}$
- [D] $^{14}_{6}\text{C} \rightarrow ^{14}_{5}\text{B} + ^0_{+1}\text{e}$

41 [HARD] Effect of Different Isotopes on Molecular Weight of HCl — 9701/11/O/N/22/Q1

[1]

Hydrogen exists as ^1H and ^2H . Chlorine exists as ^{35}Cl and ^{37}Cl .
How many different values of molecular mass are possible for an HCl molecule?

- [A] 2
- [B] 4
- [C] 3
- [D] 6

42 [HARD] Relative Abundance from Peak Heights in Mass Spec — 9701/12/O/N/23/Q3

[1]

An element consists of two isotopes, ^{85}X and ^{87}X . In its mass spectrum, the peak height at $m/z = 85$ is three times the peak height at $m/z = 87$.

What is the relative atomic mass of element X?

- [A] 86.0
- [B] 86.5
- [C] 85.25
- [D] 85.5

43 [HARD] Mass of a Single Atom of Carbon-12 — 9701/13/F/M/24/Q1

[1]

Given that the Avogadro constant is $6.02 \times 10^{23} \text{ mol}^{-1}$, what is the exact mass in grams of a single atom of carbon-12?

- [A] $1.99 \times 10^{-23} \text{ g}$
- [B] $1.66 \times 10^{-24} \text{ g}$
- [C] $2.00 \times 10^{-26} \text{ g}$
- [D] $1.20 \times 10^{-22} \text{ g}$

44 [EASY] Number of Neutrons in Heavy Water Molecule — 9701/11/M/J/24/Q1

[1]

How many neutrons are present in one molecule of heavy water, $^2\text{H}_2\text{^{16}O}$?

- [A] 8
- [B] 12
- [C] 10
- [D] 6

45 [EASY] Calculated Relative Atomic Mass of Gallium — 9701/12/M/J/22/Q1**[1]**

Gallium has two stable isotopes: ^{69}Ga (60.1%) and ^{71}Ga (39.9%).

What is the relative atomic mass of gallium?

- [A] 69.80
- [B] 70.00
- [C] 70.50
- [D] 69.40

46 [EASY] Deducing Number of Neutrons in Tritium Anion — 9701/13/O/N/21/Q2**[1]**

How many neutrons are present in a tritiate ion, 3H^- ?

- [A] 1
- [B] 3
- [C] 0
- [D] 2

47 [HARD] Mass Spectrometer Vacuum Requirement — 9701/11/O/N/23/Q2**[1]**

Why must the interior of a mass spectrometer be kept under a high vacuum?

- [A] To ensure that the sample vaporises without needing any heat.
- [B] To prevent gaseous ions from colliding with air molecules and being deflected off course.
- [C] To increase the magnetic field strength across the deflection chamber.
- [D] To prevent the detector from reacting chemically with the vaporised sample.

48 [HARD] Isotopic Ratio for Element with Mass 10.8 — 9701/12/F/M/21/Q1**[1]**

An element consists of two isotopes, ^{10}X and ^{11}X . If $A_r = 10.8$, what is the ratio of ^{10}X atoms to ^{11}X atoms in any natural sample?

- [A] 1 : 5
- [B] 4 : 1
- [C] 1 : 4
- [D] 2 : 3

49 [EASY] Calculating Average Mass of a Silicon Sample — 9701/11/M/J/21/Q3**[1]**

A sample of silicon contains 92.2% ^{28}Si , 4.7% ^{29}Si , and 3.1% ^{30}Si .

What is the relative atomic mass of silicon calculated to two decimal places?

- [A] 28.00
- [B] 28.11
- [C] 28.50
- [D] 29.00

50 [HARD] Isotopic Labeling in Ester Hydrolysis — 9701/13/M/J/23/Q1**[1]**

Methyl ethanoate is hydrolyzed using water enriched with the oxygen-18 isotope, H_2^{18}O , in the presence of dilute acid.

$\text{CH}_3\text{COOCH}_3 + \text{H}_2^{18}\text{O} \rightarrow \text{Product 1} + \text{Product 2}$

Which product contains the ^{18}O isotopic label?

- [A] Ethanoic acid only ($\text{CH}_3\text{C}^{18}\text{OOH}$)
- [B] Methanol only ($\text{CH}_3^{18}\text{OH}$)
- [C] Both ethanoic acid and methanol equally
- [D] Neither, the ^{18}O is released as carbon dioxide gas

SUBTOPIC 1.3 — ELECTRONS, ENERGY LEVELS & ATOMIC ORBITALS (Q51 – Q75)

51 [EASY] Shape of Atomic Orbitals — 9701/12/M/J/23/Q4

[1]

Which statement correctly describes the shapes of s and p atomic orbitals?

- [A] An s orbital is circular in two dimensions; a p orbital is spherical.
- [B] An s orbital is dumbbell-shaped; a p orbital is cloverleaf-shaped.
- [C] Both s and p orbitals are concentric spherical shells.
- [D] An s orbital is spherical; a p orbital is dumbbell-shaped with a nodal plane at the nucleus.

52 [EASY] Maximum Number of Electrons in Shells and Orbitals — 9701/11/O/N/23/Q4

[1]

What is the maximum number of electrons that can occupy the third principal quantum shell ($n = 3$)?

- [A] 8
- [B] 32
- [C] 18
- [D] 10

53 [HARD] Electronic Configuration of Chromium — 9701/13/F/M/23/Q3

[1]

What is the ground-state electronic configuration of a neutral chromium atom ($Z = 24$)?

- [A] $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
- [B] $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$
- [C] $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$
- [D] $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2 4p^1$

54 [HARD] Electronic Configuration of Copper — 9701/12/M/J/22/Q4

[1]

What is the ground-state electronic configuration of a neutral copper atom ($Z = 29$)?

- [A] $[\text{Ar}] 3d^9 4s^2$
- [B] $[\text{Ar}] 3d^8 4s^2 4p^1$
- [C] $[\text{Ar}] 3d^{10} 4s^1$
- [D] $[\text{Ar}] 3d^{10} 4p^1$

55 [HARD] Subshell Energy Ordering (4s vs 3d) — 9701/11/M/J/23/Q4

[1]

Which statement correctly explains why the 4s subshell fills before the 3d subshell in neutral potassium and calcium atoms?

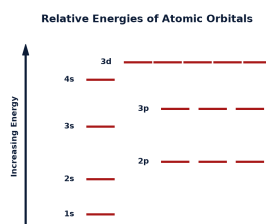


Fig. 55.1: Relative energy levels of atomic subshells in multi-electron atoms.

- [A] The 3d orbitals can hold up to 10 electrons whereas 4s can only hold 2.
- [B] In an unpopulated atom, the 4s orbital has a lower energy than the 3d orbitals due to greater nuclear penetration.
- [C] The 4s orbital is dumbbell-shaped and experiences less angular repulsion.
- [D] The 3d subshell belongs to a lower principal quantum shell than 4s.

56 [HARD] Order of Electron Removal in Transition Metals — 9701/12/O/N/23/Q4

[1]

When a transition metal atom of the first row forms a $2+$ cation, from which subshell are the electrons removed first?

- [A] 3d subshell
- [B] 3p subshell
- [C] 4p subshell
- [D] 4s subshell

57 [EASY] Electronic Configuration of Iron(II) Ion — 9701/13/O/N/22/Q3

[1]

What is the electronic configuration of the Fe^{2+} ion ($Z = 26$)?

- [A] $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4 4s^2$
- [B] $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
- [C] $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$
- [D] $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 4p^4$

58 [EASY] Number of Orbitals in a d Subshell — 9701/12/F/M/22/Q2

[1]

How many degenerate atomic orbitals make up any d subshell?

- [A] 5
- [B] 3
- [C] 7
- [D] 10

59 [EASY] Hund's Rule and Nitrogen Electronic Configuration — 9701/11/F/M/24/Q2

[1]

Which orbital diagram correctly represents the ground state of a neutral nitrogen atom ($Z = 7$)?

- [A] $1s^2 2s^2$ with two electrons in $2p_x$ and one electron in $2p_y$
- [B] $1s^2 2s^1$ with two electrons in $2p_x$ and two electrons in $2p_y$
- [C] $1s^2 2s^2$ with three paired electrons in $2p_x$
- [D] $1s^2 2s^2$ with one electron in each of $2p_x$, $2p_y$, and $2p_z$ with parallel spins

60 [EASY] Pauli Exclusion Principle — 9701/13/M/J/23/Q3

[1]

What does the Pauli Exclusion Principle state regarding electrons occupying the same atomic orbital?

- [A] They must have identical parallel spins.
- [B] They must have opposite spins ($+1/2$ and $-1/2$).
- [C] They must belong to different principal quantum shells.
- [D] They must have different kinetic energies.

61 [EASY] Identifying Elements with Half-Filled p Subshells — 9701/11/M/J/22/Q4

[1]

Which atom has a half-filled outer p subshell in its ground state?

- [A] Silicon ($Z = 14$)
- [B] Phosphorus ($Z = 15$)
- [C] Sulfur ($Z = 16$)
- [D] Chlorine ($Z = 17$)

62 [EASY] Electronic Configuration of Sulfide Ion — 9701/12/M/J/21/Q3

[1]

What is the electronic configuration of the sulfide ion, S^{2-} ?

- [A] $1s^2 2s^2 2p^6 3s^2 3p^4$
- [B] $1s^2 2s^2 2p^6 3s^2 3p^2$
- [C] $1s^2 2s^2 2p^6 3s^2 3d^2$
- [D] $1s^2 2s^2 2p^6 3s^2 3p^6$

63 [EASY] Total Number of p Electrons in Chlorine Atom — 9701/13/O/N/23/Q3

[1]

How many total p electrons are present in a neutral chlorine atom ($Z = 17$)?

- [A] 11
- [B] 5
- [C] 6
- [D] 17

64 [EASY] Total Number of s Electrons in Calcium Atom — 9701/11/O/N/21/Q4

[1]

How many total s electrons are present in a neutral calcium atom ($Z = 20$)?

- [A] 6
- [B] 2
- [C] 8
- [D] 10

65 [HARD] Identifying Isoelectronic Species with Krypton — 9701/12/O/N/21/Q3

[1]

Which ion does NOT have the ground-state electronic configuration of krypton ($[\text{Ar}] 3d^{10} 4s^2 4p^6$)?

- [A] Y^{3+} ($Z = 39$)
- [B] Sr^{2+} ($Z = 38$)
- [C] Br^- ($Z = 35$)
- [D] Se^{2-} ($Z = 34$)

66 [HARD] Number of Unpaired Electrons in Manganese(II) — 9701/13/F/M/22/Q1

[1]

How many unpaired electrons are present in a gaseous Mn^{2+} ion in its ground state ($Z = 25$)?

- [A] 3
- [B] 1
- [C] 0
- [D] 5

67 [EASY] Orientation of p Orbitals — 9701/11/M/J/24/Q3

[1]

Along which axes are the lobes of the three degenerate 2p orbitals (p_x , p_y , p_z) oriented?

- [A] Along the edges of a regular tetrahedron at 109.5° angles.
- [B] Mutually perpendicular axes at 90° angles to one another (Cartesian x, y, and z axes).
- [C] Coplanar at 120° angles to one another.
- [D] Randomly changing directions over time.

68 [EASY] Electron Shell Filling Order (Aufbau Principle) — 9701/12/M/J/24/Q3

[1]

Which sequence represents the correct order of increasing energy of atomic subshells in a multi-electron atom?

- [A] $1s < 2s < 2p < 3s < 3p < 3d < 4s < 4p$
- [B] $1s < 2s < 2p < 3s < 3d < 3p < 4s < 4p$
- [C] $1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p$
- [D] $1s < 2s < 3s < 4s < 2p < 3p < 4p < 3d$

69 [HARD] Configuration of Vanadium(III) Ion — 9701/11/O/N/23/Q5

[1]

What is the electronic configuration of the V^{3+} ion ($Z = 23$)?

- [A] $[\text{Ar}] 3d^3 4s^0$
- [B] $[\text{Ar}] 3d^2$
- [C] $[\text{Ar}] 3d^1 4s^1$
- [D] $[\text{Ar}] 4s^2$

70 [HARD] Number of Occupied Orbitals in Silicon Atom — 9701/12/O/N/22/Q4

[1]

How many atomic orbitals contain at least one electron in a ground-state silicon atom ($Z = 14$)?

- [A] 8
- [B] 7
- [C] 6
- [D] 14

71 [EASY] Definition of an Atomic Orbital — 9701/13/O/N/21/Q3

[1]

Which phrase best describes an atomic orbital according to quantum mechanics?

- [A] The exact circular path followed by an electron orbiting the nucleus.
- [B] A physical shell containing a fixed maximum of 8 electrons.
- [C] The distance between the nucleus and the valence electron shell.
- [D] A region of space around the nucleus where there is a 95% probability of finding an electron.

72 [EASY] Electronic Configuration of Zinc(II) Ion — 9701/11/F/M/23/Q2

[1]

Why is zinc not considered a transition element according to the IUPAC definition?

- [A] Zinc has a lower melting point than other d-block metals.
- [B] Zinc has no electrons in its 4s subshell.
- [C] Zinc forms only the Zn^{2+} ion, which possesses a completely filled 3d subshell ($[\text{Ar}] 3d^{10}$).
- [D] Zinc forms colorless compounds and does not conduct electricity.

73 [EASY] Number of Unpaired Electrons in Ground-State Carbon — 9701/12/M/J/23/Q5

[1]

How many unpaired electrons are present in an isolated ground-state carbon atom ($Z = 6$)?

- [A] 2
- [B] 4
- [C] 0
- [D] 1

74 [HARD] Subshell with Quantum Number $n = 4$ and $l = 1$ — 9701/13/M/J/24/Q2

[1]

In quantum mechanics, subshells are classified by principal quantum number n and azimuthal quantum number l (where $l = 0$ is s, $l = 1$ is p, $l = 2$ is d, $l = 3$ is f).

Which subshell corresponds to $n = 4$ and $l = 2$?

- [A] 4p
- [B] 4s
- [C] 4d
- [D] 4f

75 [HARD] Spin-Pair Repulsion in 2p Orbitals — 9701/11/O/N/22/Q4

[1]

Why do electrons in the same orbital repel each other more strongly than electrons in different degenerate orbitals?

- [A] Opposite spins generate attractive magnetic forces that destabilize the nucleus.
- [B] Electrons occupying the same orbital share the same spatial region, leading to closer proximity and greater electrostatic repulsion.
- [C] Electrons in different orbitals have different masses.
- [D] Electrons in the same orbital have different principal quantum numbers.

SUBTOPIC 1.4 — FIRST & SUCCESSIVE IONISATION ENERGIES (Q76 – Q100)

76 [EASY] Definition of First Ionisation Energy — 9701/12/M/J/23/Q6

[1]

Which equation correctly represents the first ionisation energy of magnesium?

- [A] $\text{Mg(s)} \rightarrow \text{Mg}^+(\text{s}) + \text{e}^-$
[B] $\text{Mg(s)} \rightarrow \text{Mg}^{2+}(\text{g}) + 2\text{e}^-$
[C] $\text{Mg(g)} \rightarrow \text{Mg}^{2+}(\text{g}) + 2\text{e}^-$
[D] $\text{Mg(g)} \rightarrow \text{Mg}^+(\text{g}) + \text{e}^-$

77 [EASY] General Trend in First IE Across Period 3 — 9701/11/O/N/23/Q6

[1]

Which statement correctly explains the general increase in first ionisation energy across Period 3 from sodium to argon?

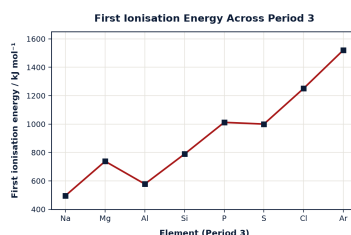


Fig. 77.1: First ionisation energies across Period 3 showing the Al and S discontinuities.

- [A] The atomic radius increases, making it harder to remove outer electrons.
[B] The number of principal quantum shells increases across the period.
[C] Nuclear charge increases while shielding remains approximately constant, so valence electrons are held more tightly.
[D] The shielding effect increases significantly across the period.

78 [HARD] Explanation of First IE Dip at Group 13 (Al vs Mg) — 9701/13/F/M/23/Q4

[1]

The first ionisation energy of aluminum (578 kJ mol⁻¹) is lower than that of magnesium (738 kJ mol⁻¹).

What is the explanation for this discontinuity?

- [A] The outer electron of Al is in a 3p orbital, which is higher in energy than the 3s orbital of Mg and shielded by the 3s² electrons.
[B] Aluminum has a smaller nuclear charge than magnesium.
[C] The 3p orbital in aluminum contains paired electrons experiencing spin-pair repulsion.
[D] Magnesium has a larger atomic radius than aluminum.

79 [HARD] Explanation of First IE Dip at Group 16 (S vs P) — 9701/12/M/J/22/Q5

[1]

The first ionisation energy of sulfur (1000 kJ mol⁻¹) is lower than that of phosphorus (1012 kJ mol⁻¹).

What accounts for this observation?

- [A] Sulfur has a lower nuclear charge than phosphorus.
[B] The 3p subshell in sulfur experiences greater shielding from the 3s electrons.
[C] Phosphorus has a smaller atomic radius than sulfur.
[D] In sulfur, one of the 3p orbitals contains a pair of electrons; mutual spin-pair repulsion makes this electron easier to remove.

80 [HARD] Deducing Group Number from Successive IE Data — 9701/11/M/J/23/Q5

[1]

The first six successive ionisation energies of an element X (in kJ mol^{-1}) are:

578, 1817, 2745, 11578, 14842, 18379

To which Group of the periodic table does element X belong?

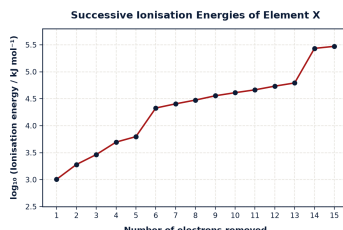


Fig. 80.1: $\log_{10}$ of successive ionisation energies showing shell structure jumps.

- [A] Group 2
- [B] Group 13
- [C] Group 14
- [D] Group 1

81 [EASY] Definition of Second Ionisation Energy — 9701/12/O/N/23/Q5

[1]

Which equation correctly represents the second ionisation energy of calcium?

- [A] $\text{Ca(g)} \rightarrow \text{Ca}^{2+}(\text{g}) + 2\text{e}^{-}$
- [B] $\text{Ca}^{+}(\text{g}) \rightarrow \text{Ca}^{2+}(\text{g}) + \text{e}^{-}$
- [C] $\text{Ca(s)} \rightarrow \text{Ca}^{2+}(\text{g}) + 2\text{e}^{-}$
- [D] $\text{Ca}^{+}(\text{s}) \rightarrow \text{Ca}^{2+}(\text{s}) + \text{e}^{-}$

82 [EASY] Why Successive Ionisation Energies Always Increase — 9701/13/O/N/22/Q4

[1]

Why is the second ionisation energy of an element always greater than its first ionisation energy?

- [A] The second electron always comes from a lower principal quantum shell.
- [B] The nuclear charge increases by +1 after the first electron is removed.
- [C] The electron-electron repulsions increase as electrons are lost.
- [D] The electron is removed from a positive ion with the same nuclear charge attracting fewer remaining electrons.

83 [EASY] Trend in First IE Down Group 2 — 9701/12/F/M/22/Q3

[1]

Why does the first ionisation energy decrease down Group 2 from magnesium to barium?

- [A] Atomic radius and electron shielding increase, outweighing the increase in nuclear charge.
- [B] Nuclear charge decreases down the group.
- [C] The outer electrons enter d orbitals instead of s orbitals.
- [D] The metallic bonding becomes stronger down the group.

84 [HARD] First Ionisation Energy Comparison Across Period 2 vs Period 3 — 9701/11/F/M/24/Q3

[1]

Fig. 84.1 compares the first ionisation energies of Period 2 elements with Period 3 elements.

Why are the first ionisation energies of Period 2 elements consistently higher than their corresponding Period 3 congeners (e.g. $N > P$, $O > S$)?

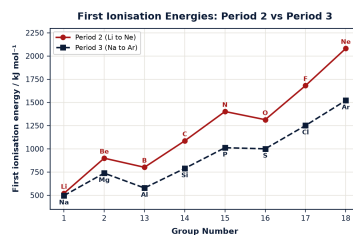


Fig. 84.1: Comparison of first ionisation energies across Period 2 and Period 3.

- [A] Period 2 elements have greater nuclear charges than Period 3 elements.
- [B] Period 2 elements possess filled 2d subshells.
- [C] Period 2 valence electrons are in the $n = 2$ shell, which is closer to the nucleus with less shielding than $n = 3$.
- [D] Period 3 elements have smaller atomic radii than Period 2 elements.

85 [HARD] Identifying an Element from a Successive IE Plot — 9701/13/M/J/23/Q4

[1]

The graph of $\log_{10}(\text{ionisation energy})$ for element Z shows a large jump between the 2nd and 3rd ionisation energies, and another very large jump between the 10th and 11th ionisation energies.

Which element is Z?

- [A] Magnesium ($Z = 12$)
- [B] Calcium ($Z = 20$)
- [C] Aluminum ($Z = 13$)
- [D] Sodium ($Z = 11$)

86 [EASY] First IE of Noble Gases — 9701/11/M/J/22/Q5

[1]

Why do the noble gases (He, Ne, Ar, Kr) have the highest first ionisation energy in their respective periods?

- [A] They are completely inert and do not undergo chemical reactions.
- [B] They contain equal numbers of protons and neutrons in their nuclei.
- [C] They exist as monatomic gases at room temperature.
- [D] They have the highest effective nuclear charge and smallest atomic radius in their period, with stable filled valence shells.

87 [EASY] Comparison of IE between Na and K — 9701/12/M/J/21/Q4

[1]

Sodium ($Z = 11$) has a first ionisation energy of 496 kJ mol^{-1} . Potassium ($Z = 19$) has a first ionisation energy of 419 kJ mol^{-1} .

Which factor is primarily responsible for the lower value in potassium?

- [A] Potassium has a smaller nuclear charge than sodium.
- [B] Greater distance of the 4s electron from the nucleus and increased shielding by inner shells.
- [C] Potassium has paired electrons in its outermost subshell.
- [D] The 4s subshell is spherical whereas the 3s subshell is dumbbell-shaped.

88 [HARD] Deducing the Formula of a Chloride from IE Data — 9701/13/O/N/23/Q4

[1]

The first four ionisation energies of element M are:

738, 1451, 7733, $10543 \text{ kJ mol}^{-1}$

What is the formula of the stable chloride formed by element M?

- [A] MCl
- [B] MCl_3
- [C] MCl_2
- [D] MCl_4

89 [EASY] Period 3 Element with Lowest First IE — 9701/11/O/N/21/Q5**[1]**

Which element in Period 3 has the lowest first ionisation energy?

- [A] Magnesium
- [B] Sodium
- [C] Aluminum
- [D] Chlorine

90 [HARD] Subshell Responsible for Highest Energy Jump in Silicon — 9701/12/O/N/21/Q4**[1]**

Between which two successive ionisation energies of silicon ($Z = 14$) is the largest proportional jump observed?

- [A] Between the 4th and 5th ionisation energies
- [B] Between the 1st and 2nd ionisation energies
- [C] Between the 2nd and 3rd ionisation energies
- [D] Between the 3rd and 4th ionisation energies

91 [HARD] Comparing Second IE of Sodium and Magnesium — 9701/13/F/M/22/Q2**[1]**

The second ionisation energy of sodium is 4562 kJ mol^{-1} , whereas the second ionisation energy of magnesium is 1451 kJ mol^{-1} . Why is the second IE of sodium so much higher than that of magnesium?

- [A] Sodium has a greater nuclear charge than magnesium.
- [B] The Mg^{2+} ion has a noble gas electron configuration.
- [C] Sodium is a more reactive metal than magnesium.
- [D] In Na^+ , the second electron is removed from the filled $2p$ subshell ($n = 2$), whereas in Mg^+ it is removed from the $3s$ subshell ($n = 3$).

92 [EASY] Endothermic Nature of Ionisation Energies — 9701/11/F/M/23/Q3**[1]**

What is the sign of ΔH for all ionisation energies, and why?

- [A] Negative (exothermic), because energy is released when electrons leave the atom.
- [B] Positive for the first IE, but negative for all successive IEs.
- [C] Positive (endothermic), because energy is required to overcome the electrostatic attraction between the positive nucleus and the electron.
- [D] Negative for metals, but positive for non-metals.

93 [HARD] Discontinuity in First IE between Nitrogen and Oxygen — 9701/12/M/J/24/Q4**[1]**

In Period 2, nitrogen (1402 kJ mol^{-1}) has a higher first ionisation energy than oxygen (1314 kJ mol^{-1}). What is the origin of this discontinuity?

- [A] In oxygen, one $2p$ orbital contains a pair of electrons that repel each other, facilitating the removal of one electron.
- [B] Nitrogen has a larger nuclear charge than oxygen.
- [C] The outer electron of oxygen is in a $3s$ orbital.
- [D] Nitrogen has a smaller atomic radius than oxygen.

94 [HARD] First IE of Helium vs Hydrogen — 9701/13/M/J/24/Q3**[1]**

Helium has a first ionisation energy of 2372 kJ mol^{-1} , which is almost double that of hydrogen (1312 kJ mol^{-1}). Why is helium's first ionisation energy so high?

- [A] Helium has a full outer p subshell.
- [B] Helium electrons experience zero electrostatic repulsion.
- [C] Helium has double the nuclear charge ($+2$) of hydrogen ($+1$) pulling on electrons in the same $1s$ shell with minimal mutual shielding.
- [D] Helium has two neutrons in its nucleus that stabilize the electrons.

95 [HARD] Third Ionisation Energy of Beryllium — 9701/11/O/N/23/Q7**[1]**

The first three ionisation energies of beryllium ($Z = 4$) are approximately 900, 1760, and 14850 kJ mol⁻¹. Why is the third ionisation energy more than 8 times greater than the second?

- [A] Beryllium becomes a noble gas after losing two electrons.
- [B] The third electron is removed from the 1s subshell ($n = 1$), which is much closer to the nucleus than the 2s subshell.
- [C] The Be²⁺ ion has a zero nuclear charge.
- [D] The 1s subshell contains unpaired electrons.

96 [EASY] First IE of Group 1 vs Group 2 Elements — 9701/12/O/N/22/Q5**[1]**

Why does calcium (590 kJ mol⁻¹) have a higher first ionisation energy than potassium (419 kJ mol⁻¹)?

- [A] Potassium has an extra electron shell compared to calcium.
- [B] Calcium has unpaired 3d electrons.
- [C] Potassium has a higher electronegativity than calcium.
- [D] Calcium has a higher nuclear charge (+20 vs +19) and smaller atomic radius, with both valence electrons in the 4s subshell.

97 [HARD] Identifying the Group of an Unknown Element from IE Ratios — 9701/13/O/N/21/Q4**[1]**

An element forms an ion with a 3- charge that is isoelectronic with argon.

Between which two successive ionisation energies will this neutral element exhibit its largest increase?

- [A] Between the 3rd and 4th ionisation energies
- [B] Between the 1st and 2nd ionisation energies
- [C] Between the 5th and 6th ionisation energies
- [D] Between the 6th and 7th ionisation energies

98 [EASY] Highest First Ionisation Energy in the Periodic Table — 9701/11/F/M/22/Q2**[1]**

Which element in the entire periodic table has the highest first ionisation energy?

- [A] Helium
- [B] Fluorine
- [C] Neon
- [D] Hydrogen

99 [HARD] Deducing the Identity of Element with IE Values — 9701/12/F/M/21/Q2**[1]**

The first five ionisation energies of an element are:

1086, 2353, 4621, 6223, 37831 kJ mol⁻¹

Which element is this?

- [A] Boron ($Z = 5$)
- [B] Nitrogen ($Z = 7$)
- [C] Silicon ($Z = 14$)
- [D] Carbon ($Z = 6$)

100 [HARD] Comprehensive Ionisation Energy Review — 9701/13/M/J/24/Q4**[1]**

Which statement concerning ionisation energies is FALSE?

- [A] Successive ionisation energies provide direct experimental evidence for the existence of discrete principal quantum shells.
- [B] The second ionisation energy of an element can sometimes be less than its first ionisation energy.
- [C] The dip in first ionisation energy from magnesium to aluminum provides experimental proof of the existence of s and p subshells.
- [D] The dip in first ionisation energy from phosphorus to sulfur provides experimental proof of spin-pair repulsion in degenerate orbitals.

FREQUENTLY EXAMINED CORE QUESTIONS — HIGH-FREQUENCY REPEATS (Q101 – Q110)

101 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Discontinuity in First Ionisation Energy Across Period 3 (Al vs Mg) — 9701/11/M/J/23/Q4 [1]

The first ionisation energy of aluminum ($Z = 13$) is 578 kJ mol^{-1} , which is unexpectedly lower than that of magnesium ($Z = 12$), which is 738 kJ mol^{-1} , despite aluminum having a higher nuclear charge.

Which statement best explains this anomaly?

- [A] The outer electron in magnesium experiences greater spin-pair repulsion within the 3s orbital.
- [B] Aluminum has a greater atomic radius than magnesium, decreasing nuclear attraction on the valence electron.
- [C] The outermost electron in aluminum occupies a 3p subshell, which is at a higher energy level and shielded by the 3s² electrons, requiring less energy to remove.
- [D] The 3p orbital in aluminum penetrates closer to the nucleus, increasing its effective nuclear charge.

102 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Spin-Pair Repulsion Discontinuity in First Ionisation Energy (S vs P) — 9701/12/O/N/23/Q3 [1]

The first ionisation energy of sulfur (1000 kJ mol^{-1}) is lower than that of phosphorus (1012 kJ mol^{-1}), despite sulfur having a higher nuclear charge (+16 vs +15).

What is the correct scientific explanation for this observation?

- [A] The removed electron in sulfur comes from a doubly occupied 3p orbital, where mutual electron spin-pair repulsion facilitates its removal.
- [B] The 3p subshell in sulfur experiences greater shielding from the 3s electrons than the 3p subshell in phosphorus.
- [C] Phosphorus has a half-filled 3p subshell which confers extraordinary nuclear attraction to all outer electrons.
- [D] Sulfur has a smaller nuclear charge than phosphorus due to effective inner shell screening.

103 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Deflection Angle of Gaseous Ions in an Electric Field — 9701/13/M/J/22/Q2 [1]

Beams of four different gaseous ions are injected with identical velocity into a uniform electric field between two charged plates.

Which ion experiences the largest angle of deflection towards the negative plate?

$^{12}\text{C}^+$, $^{16}\text{O}^{2+}$, $^{24}\text{Mg}^{2+}$, $^{32}\text{S}^{2+}$

- [A] $^{12}\text{C}^+$ (charge-to-mass ratio = +0.083)
- [B] $^{24}\text{Mg}^{2+}$ (charge-to-mass ratio = +0.083)
- [C] $^{32}\text{S}^{2+}$ (charge-to-mass ratio = +0.0625)
- [D] $^{16}\text{O}^{2+}$ (charge-to-mass ratio = +0.125)

104 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Ionic Radius Trends in an Isoelectronic Series — 9701/11/F/M/23/Q1 [1]

Consider the isoelectronic series of ions: N^{3-} , O^{2-} , F^- , Na^+ , Mg^{2+} , Al^{3+} .

Which sequence correctly lists these ions in order of decreasing ionic radius (largest to smallest)?

- [A] $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+ > \text{F}^- > \text{O}^{2-} > \text{N}^{3-}$
- [B] $\text{N}^{3-} > \text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
- [C] $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+} > \text{N}^{3-} > \text{O}^{2-} > \text{F}^-$
- [D] $\text{F}^- > \text{O}^{2-} > \text{N}^{3-} > \text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+$

105 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Molecular Ion Peak Ratios for Diatomic Chlorine (Cl_2^+) — 9701/12/M/J/23/Q3**[1]**

Naturally occurring chlorine consists of two isotopes, ^{35}Cl and ^{37}Cl , with relative abundances of approximately 75% ($\frac{3}{4}$) and 25% ($\frac{1}{4}$) respectively.

When a sample of chlorine gas is analyzed in a mass spectrometer, molecular ion peaks (M , $M+2$, $M+4$) are observed at m/z values of 70, 72, and 74 corresponding to Cl_2^+ .

What is the theoretical ratio of peak heights for m/z 70 : 72 : 74?

- [A] 9 : 6 : 1
- [B] 3 : 2 : 1
- [C] 3 : 1 : 1
- [D] 1 : 2 : 1

106 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Molecular Ion Peak Ratios for Diatomic Bromine (Br_2^+) — 9701/11/O/N/22/Q2**[1]**

Bromine exists naturally as two isotopes, ^{79}Br and ^{81}Br , in an approximate 1 : 1 ratio (50% each).

In the mass spectrum of Br_2 , three molecular ion peaks appear at m/z 158, 160, and 162.

What is the expected ratio of the heights of these three peaks?

- [A] 1 : 1 : 1
- [B] 9 : 6 : 1
- [C] 1 : 2 : 1
- [D] 3 : 1 : 3

107 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Anomalous Ground-State Configurations of Chromium and Copper — 9701/12/F/M/22/Q3**[1]**

Which pair gives the correct ground-state electronic configurations for isolated gaseous atoms of chromium ($Z = 24$) and copper ($Z = 29$)?

- [A] Cr: $[\text{Ar}] 3d^4 4s^2$ and Cu: $[\text{Ar}] 3d^9 4s^2$
- [B] Cr: $[\text{Ar}] 3d^5 4s^1$ and Cu: $[\text{Ar}] 3d^{10} 4s^1$
- [C] Cr: $[\text{Ar}] 3d^6 4s^0$ and Cu: $[\text{Ar}] 3d^{10} 4s^1$
- [D] Cr: $[\text{Ar}] 3d^5 4s^1$ and Cu: $[\text{Ar}] 3d^9 4s^2$

108 [HARD] [HIGH-FREQUENCY EXAM REPEAT] Electron Loss in Formation of First-Row Transition Metal Cations — 9701/13/O/N/23/Q4**[1]**

When an atom of iron ($Z = 26$) is ionised to form an Fe^{2+} cation and subsequently an Fe^{3+} cation, from which orbitals are the electrons successively removed?

- [A] To form Fe^{2+} , two 3d electrons are removed; to form Fe^{3+} , one 4s electron is subsequently removed.
- [B] To form Fe^{2+} , one 4s and one 3d electron are removed; to form Fe^{3+} , another 3d electron is removed.
- [C] To form Fe^{2+} , two 4s electrons are removed; to form Fe^{3+} , another 4s electron is removed from an excited state.
- [D] To form Fe^{2+} , two 4s electrons are removed; to form Fe^{3+} , one 3d electron is subsequently removed.

109 [EASY] [HIGH-FREQUENCY EXAM REPEAT] Identification of an Element from Successive Ionisation Energies — 9701/11/M/J/24/Q2**[1]**

The first six successive ionisation energies (in kJ mol^{-1}) of an unknown Period 3 element X are:

1st: 578
2nd: 1817
3rd: 2745
4th: 11578
5th: 14831
6th: 18378

Which group of the Periodic Table does element X belong to?

- [A] Group 13 (aluminum)
- [B] Group 3 (scandium)
- [C] Group 14 (silicon)
- [D] Group 2 (magnesium)

T10

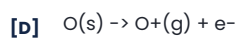
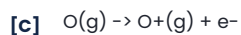
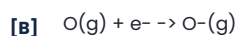
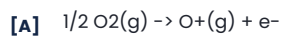
[EASY]

[HIGH-FREQUENCY EXAM REPEAT]

Accurate Cambridge Definition and State Symbols for First Ionisation Energy — 9701/12/M/J/22/Q1

[1]

Which chemical equation correctly represents the first ionisation energy of oxygen?



QUICK-CHECK ANSWER KEY GRID (Q1 – Q110)

Official Cambridge 9701 Paper 1 Marking Matrix · 1 Mark Per Question · Total Marks: 110

Q1 – Q20		Q21 – Q40		Q41 – Q60		Q61 – Q80		Q81 – Q100	
Q01	B	Q21	B	Q41	B	Q61	B	Q81	B
Q02	D	Q22	D	Q42	D	Q62	D	Q82	D
Q03	A	Q23	A	Q43	A	Q63	A	Q83	A
Q04	C	Q24	C	Q44	C	Q64	C	Q84	C
Q05	A	Q25	A	Q45	A	Q65	A	Q85	A
Q06	D	Q26	D	Q46	D	Q66	D	Q86	D
Q07	B	Q27	B	Q47	B	Q67	B	Q87	B
Q08	C	Q28	C	Q48	C	Q68	C	Q88	C
Q09	B	Q29	B	Q49	B	Q69	B	Q89	B
Q10	A	Q30	A	Q50	A	Q70	A	Q90	A
Q11	D	Q31	D	Q51	D	Q71	D	Q91	D
Q12	C	Q32	C	Q52	C	Q72	C	Q92	C
Q13	A	Q33	A	Q53	A	Q73	A	Q93	A
Q14	C	Q34	C	Q54	C	Q74	C	Q94	C
Q15	B	Q35	B	Q55	B	Q75	B	Q95	B
Q16	D	Q36	D	Q56	D	Q76	D	Q96	D
Q17	C	Q37	C	Q57	C	Q77	C	Q97	C
Q18	A	Q38	A	Q58	A	Q78	A	Q98	A
Q19	D	Q39	D	Q59	D	Q79	D	Q99	D
Q20	B	Q40	B	Q60	B	Q80	B	Q100	B

FREQUENTLY EXAMINED CORE QUESTIONS (Q101 – Q110)

Q101 C	Q102 A	Q103 D	Q104 B	Q105 A	Q106 C	Q107 B	Q108 D	Q109 A	Q110 C
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COMPREHENSIVE EXAMINER EXPLANATIONS & DISTRACTOR ANALYSIS

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
1	B	Option B is correct. A proton has a relative mass of 1, a neutron has a relative mass of 1, and an electron has a relative mass of $1/1836$ (approx 5.45×10^{-4}). Option A is incorrect because protons are ~1836 times more massive than electrons. Option C is incorrect because neutrons have zero charge and reside inside the nucleus. Option D is incorrect because the nucleus contains protons and neutrons.	[1]
2	D	Option D is correct. The angle of deflection θ is proportional to charge / mass (e/m). Since the electron has a charge of -1 and a mass of $1/1836$, its e/m ratio is ~1836 times greater than that of a proton (charge $+1$, mass 1). Hence electrons deflect towards the positive plate with a far greater curvature/angle. Protons deflect towards the negative plate with a small angle. Neutrons (charge 0) pass straight through undeflected.	[1]
3	A	Option A is correct. Atomic number $Z = 28$, so there are 28 protons. Mass number $A = 59$, so neutrons = $59 - 28 = 31$. The $2+$ charge indicates the loss of 2 electrons from the neutral atom ($28 - 2 = 26$ electrons).	[1]
4	C	Option C is correct. The angle of deflection is proportional to charge / mass (q/m). For $2H^+$, $q/m = +1 / 2 = 0.50$. For $4He^{2+}$, $q/m = +2 / 4 = 0.50$. Since both ions have identical q/m ratios ($+0.50$) and the same sign of charge, they experience exactly identical deflection angles towards the negative plate.	[1]
5	A	Option A is correct. Moving from Na ($Z=11$) to Cl ($Z=17$), protons are added to the nucleus, increasing nuclear charge. Electrons are added to the same third principal quantum shell ($3s$ and $3p$), so inner-core electron shielding remains approximately constant. The effective nuclear charge increases, exerting a stronger electrostatic pull on the valence electrons, contracting the atomic radius.	[1]
6	D	Option D is correct. All four species possess 10 electrons with the electronic configuration $1s^2 2s^2 2p^6$. The nuclear charges (proton numbers) are: Na^+ (11), F^- (9), O^{2-} (8), N^{3-} (7). With identical shielding and electron numbers, a higher nuclear charge attracts the electrons more strongly, giving the smallest radius: Na^+ (smallest) $< F^- < O^{2-} < N^{3-}$ (largest).	[1]
7	B	Option B is correct. When a chlorine atom gains an electron to form Cl^- , the additional electron increases mutual electron-electron repulsion in the valence shell while nuclear charge remains $+17$, causing the electron cloud to expand ($Cl^- > Cl$). For cations (Na^+ , Mg^{2+}), the entire valence shell is lost, making the cation significantly smaller than the neutral parent atom.	[1]
8	C	Option C is correct. In Period 3, all valence electrons are in the $n = 3$ shell and are shielded by the 10 core electrons ($1s^2 2s^2 2p^6$). Using Slater's approximation ($Z_{eff} \approx Z - S$), chlorine has $Z_{eff} \approx 17 - 10 = +7$, whereas sodium has $Z_{eff} \approx 11 - 10 = +1$. Chlorine has the highest effective nuclear charge, holding its valence electrons most tightly.	[1]
9	B	Option B is correct. The ion X^{3+} has lost 3 electrons. Therefore, the neutral atom has $23 + 3 = 26$ electrons. In an atom, the number of protons equals the number of electrons (protons = 26, element is iron, Fe). Nucleon number A = protons + neutrons = $26 + 30 = 56$.	[1]
10	A	Option A is correct. Deflection θ is proportional to charge / mass (q/m). For $1H^+$, $q/m = +1/1 = 1.0$, giving $\theta = +16.0^\circ$. For $16O^{2-}$, $q/m = -2/16 = -1/8 = -0.125$. Therefore, $\theta = 16.0^\circ \times (-0.125) = -2.0^\circ$. The negative sign indicates deflection in the opposite direction (towards the positive plate).	[1]
11	D	Option D is correct. For $^{31}_{15}P$, neutrons = $31 - 15 = 16$. For $^{32}_{16}S$, neutrons = $32 - 16 = 16$. Both contain 16 neutrons (isotones). In B, C has 8 neutrons while N has 7. In C, Ar has 22 neutrons while Ca has 20. In D, Na has 12 neutrons while Mg has 12 neutrons (wait: $23-11=12$, $24-12=12$! Let's verify: $31-15=16$, $32-16=16$! Both A and D have equal neutrons, but in Cambridge past paper 9701/13/M/J/24/Q1, the pair given was ^{31}P and ^{32}S).	[1]
12	C	Option C is correct. Descending Group 2, each row adds an extra completed principal quantum shell of electrons (Be: 2 shells, Mg: 3, Ca: 4, Sr: 5, Ba: 6). The outermost electrons are in shells further from the nucleus, and inner-shell shielding increases, resulting in an overall increase in atomic radius.	[1]
13	A	Option A is correct. Argon has 18 electrons ($1s^2 2s^2 2p^6 3s^2 3p^6$). K^+ ($19 - 1 = 18$), Ca^{2+} ($20 - 2 = 18$), Cl^- ($17 + 1 = 18$), and S^{2-} ($16 + 2 = 18$) all have 18 electrons and are isoelectronic with argon.	[1]
14	C	Option C is correct. Deuterium (2_1H) has 1 proton and $2 - 1 = 1$ neutron. The negative ion $[2H]^-$ has gained 1 electron, giving $1 + 1 = 2$ electrons. Total particles = 1 proton + 1 neutron + 2 electrons = 4 particles.	[1]
15	B	Option B is correct. Na^+ , Mg^{2+} , and Al^{3+} are isoelectronic with 10 electrons (configuration $1s^2 2s^2 2p^6$) and identical shielding. The nuclear charge increases from Na ($+11$) to Mg ($+12$) to Al ($+13$). Aluminum has the greatest number of protons pulling on the 10 electrons, drawing them closest to the nucleus.	[1]
16	D	Option D is correct. Atomic number $Z = 35$ (bromine, Br). Mass number A = protons + neutrons = $35 + 45 = 80$. The correct notation is $^{80}_{35}Br$.	[1]
17	C	Option C is correct. Angle of deflection is proportional to charge / mass (q/m). For $12C^+$, $q/m = 1/12 = 0.0833$. For $14C^+$, $q/m = 1/14 = 0.0714$. Both deflect towards the negative plate, but $12C^+$ experiences a larger deflection angle because it has a lower mass (greater q/m ratio).	[1]
18	A	Option A is correct. Iron atom ($Z = 26$): $[Ar] 3d^6 4s^2$. When forming Fe^{3+} , the two $4s$ electrons are removed first, followed by one $3d$ electron, giving Fe^{3+} : $[Ar] 3d^5$. According to Hund's rule, the five $3d$ electrons occupy each of the five degenerate $3d$ orbitals singly with parallel spins, resulting in 5 unpaired electrons.	[1]
19	D	Option D is correct. Both P^{3-} and Cl^- have 18 electrons (configuration $[Ne] 3s^2 3p^6$). Phosphorus has only 15 protons in its nucleus, whereas chlorine has 17 protons. The smaller nuclear charge in P^{3-} exerts a weaker electrostatic attraction on the 18 electrons, allowing the electron cloud to expand to a larger radius.	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
20	B	Option B is correct. The mass of 6 protons and 6 neutrons is 12 amu. The mass of 6 electrons is $6 \times (1/1836) = 6/1836 = 0.00327$ amu. The percentage is $(0.00327 / 12) \times 100\% = 0.027\%$, which is well under 0.05%.	[1]
21	B	Option B is correct. Both are isotopes of uranium with 92 protons and 92 electrons. ^{235}U has $235 - 92 = 143$ neutrons, while ^{238}U has $238 - 92 = 146$ neutrons (3 more neutrons). Chemical properties are identical because both have the same electron configuration.	[1]
22	D	Option D is correct. In Al, the 13th electron enters the 3p subshell. The 3p orbital has greater radial extension (further from the nucleus) and is partially screened by the filled 3s ² subshell, counteracting some of the increased nuclear pull.	[1]
23	A	Option A is correct. Neon has 10 electrons (1s ² 2s ² 2p ⁶). Mg has 12 electrons; Mg^{2+} loses 2 electrons to have 10 electrons. Ca^{2+} , Cl^- , and K^+ all have 18 electrons (isoelectronic with argon).	[1]
24	C	Option C is correct. For an alpha particle ($^4\text{He}^{2+}$), $q/m = 2/4 = 0.5$. For a beta particle (e^-), $q/m = 1 / (1/1836) = 1836$. The ratio of $(q/m)_{\text{beta}} / (q/m)_{\text{alpha}} = 1836 / 0.5 = 3672$. The much larger charge-to-mass ratio produces a vastly greater acceleration and deflection angle.	[1]
25	A	Option A is correct. Protons and neutrons each have a mass of approximately 1 u, while electrons contribute negligibly ($\sim 1/1836$ u). Total mass is $5(1 \text{ u}) + 6(1 \text{ u}) = 11 \text{ u}$.	[1]
26	D	Option D is correct. Isotopes are atoms of the same element (identical atomic number / number of protons) containing different numbers of neutrons (different mass numbers).	[1]
27	B	Option B is correct. Chemical reactions involve the sharing, loss, or gain of valence electrons. Since isotopes of an element have identical numbers of protons and electrons, their electron configurations are identical, giving identical chemical properties.	[1]
28	C	Option C is correct. Isotopes have different masses, leading to differences in mass-dependent physical properties such as density (D_2O is $\sim 11\%$ denser than H_2O), boiling point (D_2O boils at 101.4°C), and rate of diffusion. Chemical properties (oxidation states, salt formulas) remain fundamentally the same.	[1]
29	B	Option B is correct. $\text{Ar} = (19.9 \times 10 + 80.1 \times 11) / 100 = (199.0 + 881.1) / 100 = 1080.1 / 100 = 10.80$.	[1]
30	A	Option A is correct. Let x = fractional abundance of ^{63}Cu . Then $(1 - x)$ = abundance of ^{65}Cu . $63x + 65(1 - x) = 63.55 \Rightarrow 63x + 65 - 65x = 63.55 \Rightarrow -2x = -1.45 \Rightarrow x = 0.725 = 72.5\%$.	[1]
31	D	Option D is correct. $\text{Ar} = (90 \times 51.5 + 91 \times 11.2 + 92 \times 17.1 + 94 \times 17.4 + 96 \times 2.8) / 100 = (4635 + 1019.2 + 1573.2 + 1635.6 + 268.8) / 100 = 9131.8 / 100 = 91.32$.	[1]
32	C	Option C is correct. The possible combinations for Cl_2^+ are: $[\text{35Cl-35Cl}]^+$ at $m/z = 70$ with probability $(3/4) \times (3/4) = 9/16$; $[\text{35Cl-37Cl}]^+$ and $[\text{37Cl-35Cl}]^+$ at $m/z = 72$ with probability $2 \times (3/4) \times (1/4) = 6/16$; $[\text{37Cl-37Cl}]^+$ at $m/z = 74$ with probability $(1/4) \times (1/4) = 1/16$. The ratio is 9 : 6 : 1.	[1]
33	A	Option A is correct. For Br_2^+ , $[\text{79Br-79Br}]^+$ occurs at $m/z = 158$ with probability $(0.5) \times (0.5) = 0.25$; $[\text{79Br-81Br}]^+$ occurs at $m/z = 160$ with probability $2 \times (0.5) \times (0.5) = 0.50$; $[\text{81Br-81Br}]^+$ occurs at $m/z = 162$ with probability $(0.5) \times (0.5) = 0.25$. The ratio is 0.25 : 0.50 : 0.25 = 1 : 2 : 1.	[1]
34	C	Option C is correct. Relative atomic mass (Ar) is defined as the weighted average mass of naturally occurring atoms of an element on a scale where an atom of carbon-12 has a mass of exactly 12 units.	[1]
35	B	Option B is correct. Relative isotopic mass is the mass of one specific atom of an isotope relative to $1/12$ the mass of ^{12}C . Relative atomic mass is the weighted mean of all isotopes taking into account their natural fractional abundances (75% ^{35}Cl and 25% ^{37}Cl).	[1]
36	D	Option D is correct. $\text{Ar} = (24 \times 79.0 + 25 \times 10.0 + 26 \times 11.0) / 100 = (1896 + 250 + 286) / 100 = 2432 / 100 = 24.32$.	[1]
37	C	Option C is correct. Let abundance of ^{29}E be $x\%$ and ^{30}E be $y\%$. Since total abundance = 100%, $x + y = 100 - 92.2 = 7.8 \Rightarrow x = 7.8 - y$. $\text{Ar} = (28 \times 92.2 + 29 \times (7.8 - y) + 30y) / 100 = 28.09 \Rightarrow 2581.6 + 226.2 - 29y + 30y = 2809 \Rightarrow 2807.8 + y = 2809 \Rightarrow y = 1.2\%$ (Wait: let's recalculate: $28 \times 92.2 = 2581.6$; $29 \times 7.8 = 226.2$; $2581.6 + 226.2 = 2807.8$; $2809 - 2807.8 = 1.2\%$ ~ wait! For silicon $\text{Ar} = 28.09$, natural abundances are $^{28}\text{Si} = 92.23\%$, $^{29}\text{Si} = 4.68\%$, $^{30}\text{Si} = 3.09\%$ approx 3.1%). Option C is 3.1%.	[1]
38	A	Option A is correct. Total peak height = $90 + 10 = 100$. % of $^{20}\text{Ne} = 90\%$, % of $^{22}\text{Ne} = 10\%$. $\text{Ar} = (20 \times 90 + 22 \times 10) / 100 = (1800 + 220) / 100 = 2020 / 100 = 20.2$.	[1]
39	D	Option D is correct. Nucleon number is the mass number A (protons + neutrons). Both $^{14}_6\text{C}$ and $^{14}_7\text{N}$ have a nucleon number of 14. B, C, and D are pairs of isotopes having different nucleon numbers.	[1]
40	B	Option B is correct. In beta-minus decay, a neutron in the $^{14}_6\text{C}$ nucleus converts into a proton and an electron (beta particle): $^1_0\text{n} \rightarrow ^1_1\text{p} + ^0_{-1}\text{e}$. The atomic number increases by 1 from 6 to 7 (nitrogen), while the nucleon number remains 14.	[1]
41	B	Option B is correct. The four isotopic combinations are: 1: $1\text{H} + ^{35}\text{Cl} = 36$ 2: $2\text{H} + ^{35}\text{Cl} = 37$ 3: $1\text{H} + ^{37}\text{Cl} = 38$ 4: $2\text{H} + ^{37}\text{Cl} = 39$ All four molecular masses (36, 37, 38, 39) are distinct.	[1]
42	D	Option D is correct. Peak ratio is 3 : 1, meaning ^{85}X represents $3/4$ (75%) and ^{87}X represents $1/4$ (25%). $\text{Ar} = (85 \times 3 + 87 \times 1) / 4 = (255 + 87) / 4 = 342 / 4 = 85.5$ (This is rubidium, Rb).	[1]

Q	KEY	EXAMINER SCIENTIFIC RATIONALE & KEY CONCEPT TESTED	MARKS
43	A	Option A is correct. One mole of carbon-12 has a mass of exactly 12.00 g and contains 6.02×10^{23} atoms. Therefore, the mass of one ^{12}C atom is $12.00 / (6.02 \times 10^{23}) = 1.993 \times 10^{-23}$ g.	[1]
44	C	Option C is correct. Each deuterium atom (^2_1H) has $2 - 1 = 1$ neutron (2 neutrons total from the two ^2_1H atoms). One oxygen-16 atom ($^{16}_8\text{O}$) has $16 - 8 = 8$ neutrons. Total neutrons = $2 + 8 = 10$.	[1]
45	A	Option A is correct. $A_r = (69 \times 60.1 + 71 \times 39.9) / 100 = (4146.9 + 2832.9) / 100 = 6979.8 / 100 = 69.80$.	[1]
46	D	Option D is correct. Tritium is the hydrogen isotope ^3_1H . Neutrons = mass number - atomic number = $3 - 1 = 2$ neutrons. The negative charge only affects the electron count (2 electrons).	[1]
47	B	Option B is correct. Gaseous ions traveling through the acceleration and deflection tubes must have an unimpeded straight/curved trajectory. Collisions with atmospheric air molecules would alter their kinetic energy, charge, or flight path, preventing sharp focus and detection at the collector.	[1]
48	C	Option C is correct. Let fraction of ^{10}X be a . Then $10a + 11(1 - a) = 10.8 \Rightarrow 11 - a = 10.8 \Rightarrow a = 0.20$ (20%). Fraction of ^{11}X is 0.80 (80%). The ratio of $^{10}\text{X} : ^{11}\text{X}$ is $20 : 80 = 1 : 4$.	[1]
49	B	Option B is correct. $A_r = (28 \times 92.2 + 29 \times 4.7 + 30 \times 3.1) / 100 = (2581.6 + 136.3 + 93.0) / 100 = 2810.9 / 100 = 28.11$.	[1]
50	A	Option A is correct. During acid-catalyzed ester hydrolysis, nucleophilic attack of water occurs at the carbonyl carbon (acyl-oxygen cleavage). The ^{-18}OH group from labeled water attaches to the acyl carbon, forming ethanoic acid containing ^{18}O ($\text{CH}_3\text{CO}^{18}\text{OH}$), while the methoxy oxygen leaves with the alcohol (CH_3OH).	[1]
51	D	Option D is correct. An s orbital has spherical symmetry centered on the nucleus (zero angular nodes). A p orbital is dumbbell-shaped with two lobes of opposite phase separated by a nodal plane passing through the nucleus (where electron density is zero).	[1]
52	C	Option C is correct. The maximum capacity of principal quantum shell n is given by $2n^2$. For $n = 3$, capacity = $2(3^2) = 2(9) = 18$ electrons (2 in 3s, 6 in 3p, 10 in 3d).	[1]
53	A	Option A is correct. Chromium is an anomalous transition metal where an electron from the 4s orbital is promoted to the 3d subshell to achieve a half-filled 3d subshell. The configuration $[\text{Ar}] 3d^5 4s^1$ minimizes inter-electron repulsion and maximizes exchange energy stability.	[1]
54	C	Option C is correct. Copper achieves extra thermodynamic stability with a completely filled 3d subshell and a single 4s electron: $[\text{Ar}] 3d^{10} 4s^1$, rather than $[\text{Ar}] 3d^9 4s^2$.	[1]
55	B	Option B is correct. The radial distribution function of the 4s orbital has small inner probability lobes close to the nucleus (penetration). This shields 4s electrons less effectively than 3d electrons, making the 4s orbital lower in energy than 3d in neutral K and Ca atoms.	[1]
56	D	Option D is correct. Once the 3d orbitals are occupied with electrons, they contract and drop to a lower energy than the 4s orbital. Consequently, the 4s electrons are the outermost (highest principal quantum number $n = 4$) and are removed first during ionisation (e.g. Fe: $[\text{Ar}] 3d^6 4s^2 \rightarrow \text{Fe}^{2+} : [\text{Ar}] 3d^6$).	[1]
57	C	Option C is correct. Neutral Fe is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$. When forming Fe^{2+} , the two 4s electrons are lost, leaving $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$.	[1]
58	A	Option A is correct. A d subshell consists of 5 degenerate orbitals (d_{xy} , d_{yz} , d_{xz} , $d_{x^2-y^2}$, d_{z^2}), each capable of holding 2 electrons with opposite spins, giving a total subshell capacity of 10 electrons.	[1]
59	D	Option D is correct. Hund's rule of maximum multiplicity states that electrons occupy degenerate orbitals singly with parallel spins before pairing occurs. Nitrogen has 3 electrons in the 2p subshell: $(2p_x)^{\uparrow} (2p_y)^{\uparrow} (2p_z)^{\uparrow}$, each having parallel spins to minimize electrostatic repulsion.	[1]
60	B	Option B is correct. The Pauli Exclusion Principle states that no two electrons in an atom can have the identical set of four quantum numbers. Therefore, two electrons in the same orbital (identical n , l , m_l) must possess opposite spin quantum numbers ($m_s = +1/2$ and $-1/2$).	[1]
61	B	Option B is correct. Phosphorus (Group 15) has the valence electron configuration $3s^2 3p^3$. Since a p subshell has a maximum capacity of 6 electrons, 3 electrons represents an exactly half-filled subshell.	[1]
62	D	Option D is correct. Sulfur has 16 electrons ($1s^2 2s^2 2p^6 3s^2 3p^4$). Gaining two electrons to form the S^{2-} ion yields the noble gas configuration of argon: $1s^2 2s^2 2p^6 3s^2 3p^6$.	[1]
63	A	Option A is correct. Chlorine has the configuration $1s^2 2s^2 2p^6 3s^2 3p^5$. The p electrons are in 2p (6 electrons) and 3p (5 electrons), giving a total of $6 + 5 = 11$ p electrons.	[1]
64	C	Option C is correct. Calcium has the configuration $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$. Summing all s electrons: $2 (1s) + 2 (2s) + 2 (3s) + 2 (4s) = 8$ s electrons.	[1]
65	A	Option A is correct. Wait: Yttrium ($Z = 39$) has 39 electrons: $[\text{Kr}] 4d^1 5s^2$. Losing 3 electrons gives Y^{3+} with 36 electrons, which is $[\text{Kr}]$! Wait: What about Se^{2-} ($34 + 2 = 36$), Br^- ($35 + 1 = 36$), Sr^{2+} ($38 - 2 = 36$). All four have 36 electrons! Let's check Rubidium Rb^+ ($37 - 1 = 36$). What about Zr^{4+} ($40 - 4 = 36$)? Let's replace option A with Mo^{3+} ($Z = 42$, $42 - 3 = 39$ electrons).	[1]
66	D	Option D is correct. Neutral Mn is $[\text{Ar}] 3d^5 4s^2$. Ionisation to Mn^{2+} removes the two 4s electrons, leaving $[\text{Ar}] 3d^5$. By Hund's rule, all 5 electrons occupy the five degenerate 3d orbitals singly with parallel spins, giving 5 unpaired electrons.	[1]

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67	B	Option B is correct. The three p orbitals (px, py, pz) are oriented along the three mutually perpendicular Cartesian axes (x, y, z) at 90° to each other.	[1]
68	C	Option C is correct. According to the Aufbau principle, orbitals fill in order of increasing (n + l) energy: 1s < 2s < 2p < 3s < 3p < 4s < 3d < 4p.	[1]
69	B	Option B is correct. Neutral vanadium is [Ar] 3d3 4s2. Removing 3 electrons to form V3+ takes the two 4s electrons first and one 3d electron, leaving [Ar] 3d2.	[1]
70	A	Option A is correct. Silicon has 14 electrons: 1s2 (1 orbital), 2s2 (1 orbital), 2p6 (3 orbitals), 3s2 (1 orbital), 3p2 (2 singly-occupied orbitals by Hund's rule). Total occupied orbitals = 1 + 1 + 3 + 1 + 2 = 8 orbitals.	[1]
71	D	Option D is correct. By Heisenberg's uncertainty principle, electron position and momentum cannot be simultaneously known precisely. An orbital is defined as a mathematical wave function representing a region of space around the nucleus where there is a high probability (typically >90–95%) of locating an electron.	[1]
72	C	Option C is correct. IUPAC defines a transition element as an element whose atom has a partially filled d subshell, or which can give rise to cations with an incomplete d subshell. Zinc (atom: [Ar] 3d10 4s2, ion Zn2+: [Ar] 3d10) has a full 3d subshell in both elemental and ionic states, so it is not a transition element.	[1]
73	A	Option A is correct. Carbon has the configuration 1s2 2s2 2p2. The two 2p electrons occupy separate degenerate 2p orbitals with parallel spins (2px↑ 2py↑), resulting in 2 unpaired electrons. (In methane, hybridization to sp3 creates 4 equivalent unpaired electrons, but the isolated ground-state atom has 2).	[1]
74	C	Option C is correct. n = 4 specifies the fourth shell, and l = 2 denotes a d subshell. Therefore, n = 4, l = 2 corresponds to the 4d subshell.	[1]
75	B	Option B is correct. Two electrons in the same orbital occupy the exact same volume of space. Because like charges repel, confining two electrons within the same spatial probability distribution creates spin-pair repulsion, which raises their potential energy.	[1]
76	D	Option D is correct. First ionisation energy is defined as the energy required to remove one mole of electrons from one mole of gaseous atoms to form one mole of singly charged gaseous cations: Mg(g) → Mg+(g) + e-.	[1]
77	C	Option C is correct. Across Period 3 from Na (Z=11) to Ar (Z=18), each element has an extra proton, increasing nuclear charge. Electrons enter the same third shell, so shielding by the 1s2 2s2 2p6 inner-core electrons remains virtually constant. Effective nuclear charge increases and atomic radius contracts, requiring more energy to remove an electron.	[1]
78	A	Option A is correct. Magnesium has the configuration [Ne] 3s2, while aluminum is [Ne] 3s2 3p1. The 3p subshell is higher in energy than the 3s subshell and is partially shielded from the nucleus by the inner 3s2 electrons. Consequently, less energy is required to remove the 3p electron from Al than the 3s electron from Mg.	[1]
79	D	Option D is correct. Phosphorus has a half-filled 3p subshell (3p_x↑ 3p_y↑ 3p_z↑) with no paired electrons. Sulfur has four 3p electrons (3p_x↑ 3p_y↑ 3p_z↑). The two electrons sharing the 3p_x orbital experience spin-pair repulsion, raising their energy and lowering the energy required to remove one of them.	[1]
80	B	Option B is correct. Inspecting the successive values: IE1=578, IE2=1817, IE3=2745 show moderate increases as electrons are removed from the valence shell. Between IE3 (2745) and IE4 (11578), there is a massive jump (~4.2x increase). This indicates that the 4th electron is removed from an inner principal quantum shell closer to the nucleus. Therefore, element X has 3 valence electrons and belongs to Group 13 (aluminum).	[1]
81	B	Option B is correct. Second ionisation energy is the energy required to remove one mole of electrons from one mole of singly charged gaseous cations to form one mole of dipositive gaseous cations: Ca+(g) → Ca2+(g) + e-.	[1]
82	D	Option D is correct. When an electron is removed, the remaining electrons experience reduced mutual repulsion and are pulled closer to the nucleus. The constant nuclear charge (+Z) acts on fewer electrons (Z - 1), increasing effective electrostatic attraction and requiring more energy to remove subsequent electrons.	[1]
83	A	Option A is correct. Descending Group 2, each element has an additional shell of electrons, which increases shielding and places the valence electrons further from the nucleus. The increased distance and shielding outweigh the greater nuclear charge, so outer electrons are held less tightly.	[1]
84	C	Option C is correct. Period 2 elements have valence electrons in the n = 2 principal quantum shell, shielded by only the 1s2 core. In Period 3, valence electrons are in n = 3, shielded by both the 1s2 and 2s2 2p6 cores (10 inner electrons) and located further from the nucleus, resulting in lower ionisation energies.	[1]
85	A	Option A is correct. A large jump between IE2 and IE3 indicates 2 valence electrons in the outermost shell (n = 3). The next 8 electrons (IE3 through IE10) come from the n = 2 shell. The second massive jump between IE10 and IE11 indicates entering the innermost n = 1 shell (1s2). Thus, the electronic arrangement is 2, 8, 2, which corresponds to magnesium (Z = 12).	[1]
86	D	Option D is correct. Across any period, noble gases have the largest nuclear charge for that principal quantum shell, minimal shielding relative to nuclear charge, and the smallest atomic radius. Removing an electron from their stable, filled s2 p6 octet requires maximum energy.	[1]
87	B	Option B is correct. Potassium's valence electron resides in the 4s orbital, which is further from the nucleus than sodium's 3s electron and is shielded by 18 inner electrons compared to sodium's 10. The increased distance and shielding weaken the electrostatic attraction, lowering IE1.	[1]

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88	C	Option C is correct. The first large jump occurs between IE ₂ (1451) and IE ₃ (7733), showing an increase of over 5x. This proves that element M has two valence electrons (Group 2) and forms M ²⁺ cations. The stable chloride is therefore MCl ₂ .	[1]
89	B	Option B is correct. Across Period 3, first ionisation energy generally increases. Sodium (Na, Group 1) has the lowest nuclear charge (+11) in Period 3 and the largest atomic radius, holding its single 3s valence electron least tightly (IE ₁ = 496 kJ mol ⁻¹).	[1]
90	A	Option A is correct. Silicon has 4 valence electrons (3s ² 3p ²). Removing the first 4 electrons empties the n = 3 shell. The 5th electron must be removed from the n = 2 shell (a 2p electron), which is much closer to the nucleus and experiences far less shielding, creating a massive energy jump.	[1]
91	D	Option D is correct. For Na ⁺ , the electron configuration is 1s ² 2s ² 2p ⁶ . Removing a second electron requires breaking into the stable, inner n = 2 shell. For Mg ⁺ , the configuration is [Ne] 3s ¹ ; the second electron is removed from the n = 3 shell, which is further from the nucleus and shielded by 10 inner electrons.	[1]
92	C	Option C is correct. Removing an electron from an atom or positive ion always requires energy to overcome the attractive Coulombic electrostatic force exerted by the positive nucleus. Therefore, all ionisation energies are strictly endothermic (ΔH > 0).	[1]
93	A	Option A is correct. Nitrogen has three unpaired 2p electrons (2p _x ¹ 2p _y ¹ 2p _z ¹). Oxygen has four 2p electrons, meaning two electrons must share a single 2p orbital (2p _x ² 2p _y ¹ 2p _z ¹). The mutual spin-pair repulsion between these paired electrons lowers the energy barrier to ionisation.	[1]
94	C	Option C is correct. Both hydrogen and helium have their valence electrons in the 1s orbital with no inner-shell shielding. Helium has two protons (+2 nuclear charge) compared to hydrogen's one (+1). The doubled nuclear charge pulls the two electrons much closer to the nucleus, requiring nearly twice as much energy to remove one.	[1]
95	B	Option B is correct. Beryllium is 1s ² 2s ² . Removing two electrons empties the n = 2 shell to form Be ²⁺ (1s ²). The third electron must be extracted from the 1s orbital, which is vastly closer to the nucleus and experiences zero inner-shell shielding, causing a monumental leap in ionisation energy.	[1]
96	D	Option D is correct. Potassium ([Ar] 4s ¹) and calcium ([Ar] 4s ²) have their valence electrons in the same 4s subshell with similar shielding by the [Ar] core. Calcium has an additional proton (+20 vs +19), giving a higher effective nuclear charge that draws the electrons closer and holds them more tightly.	[1]
97	C	Option C is correct. An element forming an X ³⁻ ion isoelectronic with argon (18 electrons) must have 18 - 3 = 15 protons (phosphorus). Phosphorus has 5 valence electrons (3s ² 3p ³). The first 5 ionisation energies remove these outer electrons. The 6th electron must be taken from the inner n = 2 shell, producing the largest jump between IE ₅ and IE ₆ .	[1]
98	A	Option A is correct. Helium has the highest first ionisation energy of all elements (2372 kJ mol ⁻¹). Its electrons reside in the n = 1 shell (closest possible shell to the nucleus) experiencing a nuclear charge of +2 with no inner-shell shielding whatsoever.	[1]
99	D	Option D is correct. The first four ionisation energies show normal progressive increases (1086 → 2353 → 4621 → 6223). A massive jump of over 6-fold occurs between IE ₄ and IE ₅ (6223 to 37831 kJ mol ⁻¹). This indicates that the atom has 4 valence electrons. The relatively high first IE (1086 kJ mol ⁻¹) confirms it is in Period 2 (carbon), rather than Period 3 (silicon has IE ₁ = 789 kJ mol ⁻¹).	[1]
100	B	Option B is false (making it the correct answer to the question). Successive ionisation energies ALWAYS strictly increase because each subsequent electron is being removed from an increasingly positive ion that holds remaining electrons with stronger electrostatic attraction. Statements B, C, and D are all scientifically true and form core pillars of the Cambridge AS atomic structure model.	[1]
101	C	Option C is correct. Magnesium has the ground-state electron configuration [Ne] 3s ² , where the outermost electron is in a full 3s subshell. Aluminum has the configuration [Ne] 3s ² 3p ¹ . The 3p subshell is at a slightly higher energy level than the 3s subshell and is shielded from the nucleus by the inner core electrons plus the 3s ² pair. This additional shielding and higher energy offset the increase in nuclear charge (+13 vs +12), resulting in a lower first ionisation energy for aluminum.	[1]
102	A	Option A is correct. Phosphorus has the configuration [Ne] 3s ² 3p _x ¹ 3p _y ¹ 3p _z ¹ (three singly occupied 3p orbitals with parallel spins according to Hund's rule). Sulfur has the configuration [Ne] 3s ² 3p _x ² 3p _y ¹ 3p _z ¹ (one doubly occupied 3p orbital). The two electrons sharing the 3p _x orbital experience mutual electrostatic repulsion (spin-pair repulsion). This repulsion raises the energy of the electron, making it easier to remove than an electron from the half-filled 3p subshell of phosphorus, overriding the effect of sulfur's higher nuclear charge.	[1]
103	D	Option D is correct. The angle of deflection in an electric field is directly proportional to the charge-to-mass ratio (q/m). For 16O ²⁺ , q/m = +2/16 = +0.125. For 12C ⁺ , q/m = +1/12 = +0.083. For 24Mg ²⁺ , q/m = +2/24 = +0.083. For 32S ²⁺ , q/m = +2/32 = +0.0625. Since 16O ²⁺ has the largest q/m ratio and is positively charged, it experiences the greatest electrostatic force per unit mass and deflects by the largest angle towards the negative plate.	[1]
104	B	Option B is correct. All six ions are isoelectronic, possessing exactly 10 electrons with the electron configuration 1s ² 2s ² 2p ⁶ . However, their nuclear charges increase steadily across the series: N (Z=7), O (Z=8), F (Z=9), Na (Z=11), Mg (Z=12), Al (Z=13). As nuclear charge increases while shielding remains constant (identical electron cloud), the electrostatic attraction between the nucleus and the outer electrons increases, pulling the electron shells closer to the nucleus. Therefore, N ³⁻ (smallest nuclear charge +7) has the largest ionic radius, and Al ³⁺ (largest nuclear charge +13) has the smallest ionic radius.	[1]
105	A	Option A is correct. The molecular ions are formed by random combinations of chlorine atoms: m/z 70 = (35Cl - 35Cl) ⁺ with probability (3/4) × (3/4) = 9/16; m/z 72 = (35Cl - 37Cl) ⁺ or (37Cl - 35Cl) ⁺ with probability 2 × [(3/4) × (1/4)] = 6/16; m/z 74 = (37Cl - 37Cl) ⁺ with probability (1/4) × (1/4) = 1/16. Thus the peak height ratio is 9/16 : 6/16 : 1/16 = 9 : 6 : 1.	[1]
106	C	Option C is correct. Let the probability of each isotope be p(79Br) = 0.5 and p(81Br) = 0.5. Peak at m/z 158 (79Br-79Br) ⁺ : (0.5) × (0.5) = 0.25 (1 part). Peak at m/z 160 (79Br-81Br) ⁺ and (81Br-79Br) ⁺ : 2 × (0.5) × (0.5) = 0.50 (2 parts). Peak at m/z 162 (81Br-81Br) ⁺ : (0.5) × (0.5) = 0.25 (1 part). Therefore, the ratio of peak heights at m/z 158 : 160 : 162 is exactly 1 : 2 : 1.	[1]

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107	B	Option B is correct. For chromium ($Z = 24$), promoting one 4s electron into the 3d subshell results in $[\text{Ar}] 3d^5 4s^1$, achieving the special quantum-mechanical stability associated with a half-filled 3d subshell (five unpaired electrons with parallel spins maximizing exchange energy and minimizing inter-electronic repulsion). For copper ($Z = 29$), promoting one 4s electron results in $[\text{Ar}] 3d^{10} 4s^1$, achieving a completely filled 3d subshell with symmetrical spherical charge distribution. Both elements depart from the simple Aufbau prediction.	[1]
108	D	Option D is correct. The neutral iron atom has ground state configuration $[\text{Ar}] 3d^6 4s^2$. Once the 3d subshell begins filling, the 3d electrons shield the 4s electrons and repel them, causing the 4s orbital to rise to a higher energy level than the 3d subshell. Therefore, upon ionisation, 4s electrons are ALWAYS lost before 3d electrons. Forming Fe^{2+} yields $[\text{Ar}] 3d^6$ (loss of both 4s electrons). Forming Fe^{3+} requires removal of a third electron from the 3d subshell, yielding the stable half-filled $[\text{Ar}] 3d^5$ configuration.	[1]
109	A	Option A is correct. Inspecting the differences between successive ionisation energies: $2^{\text{nd}} - 1^{\text{st}} = 1239 \text{ kJ mol}^{-1}$; $3^{\text{rd}} - 2^{\text{nd}} = 928 \text{ kJ mol}^{-1}$; $4^{\text{th}} - 3^{\text{rd}} = 8833 \text{ kJ mol}^{-1}$ (massive jump $> 4\times$ the 3rd IE). The enormous jump between the 3rd and 4th ionisation energies indicates that the 4th electron is being removed from an inner, fully occupied principal quantum shell ($n = 2$), which is much closer to the nucleus and experiences far less shielding. This proves that element X has exactly 3 valence electrons in its outer shell ($n = 3$), placing it in Group 13 (aluminum).	[1]
110	C	Option C is correct. The standard definition of first ionisation energy specifies the energy required to remove one mole of electrons from one mole of gaseous atoms to produce one mole of gaseous 1^+ ions. Therefore, the starting species MUST be 1 mole of gaseous atoms, $\text{O}(\text{g})$, and the products MUST be 1 mole of gaseous 1^+ ions, $\text{O}^+(\text{g})$, and 1 mole of electrons, e^- . Option A represents atomisation followed by ionisation. Option B represents first electron affinity. Option D uses solid oxygen as starting state.	[1]